CG2 attributes v3 · Grounding DINO top-3, trained on Claude's labels

Mind map of what was built in this round and the measured results. Reference labels: Claude's visual labels (not human-verified); every number below is on the 240 held-out TEST objects unless stated, which were never used for training or for a choice.



Leaf colour: green = done earlier, blue = reused, orange = new in this round, red = limitation / decision.

1. Is Grounding DINO fine-tuned?

- **Approach A (in use):** Grounding DINO stays pretrained; a small layer is **trained on Claude's labels** to turn its evidence (phrase scores, boxes, box counts) plus the earlier image model into the six attributes. It runs on every object without new GPU work.
- **Test B:** Grounding DINO's **own decoder weights** fine-tuned on the same labels, the object box tagged with its class phrase.

11.2 M parameters trained (Grounding DINO decoder + box head; Swin backbone, BERT and the feature enhancer frozen) on 1,504 TRAIN objects; epoch 2 chosen on validation; 76.4 min in total.

- epoch 1: training loss 0.7735, validation mean R@1 54.0% (38.2 min)
- epoch 2: training loss 0.593, validation mean R@1 54.6% (73.2 min)

attribute	pretrained DINO R@1 / R@3	fine-tuned DINO R@1 / R@2 / R@3	approach A R@1 / R@2 / R@3
Object form	22.7% / 66.8%	73.1% / 83.2% / 88.2%	83.6% / 89.9% / 92.0%
Surface finish	6.2% / 35.4%	36.2% / 60.8% / 71.2%	45.4% / 65.8% / 75.4%
Transparency	35.4% / 90.0%	72.1% / 84.6% / 93.8%	74.6% / 92.1% / 98.8%
Surface texture	4.2% / 75.4%	85.0% / 91.7% / 96.7%	85.0% / 95.0% / 99.2%
Glaze effects	5.7% / 33.9%	39.5% / 57.3% / 74.8%	49.4% / 66.6% / 78.2%
Possible defects	18.2% / 54.5%	27.3% / 72.7% / 100.0%	50.0% / 63.6% / 77.3%

Decision (made on validation, confirmed on test): fine-tuning lifts Grounding DINO far above its pretrained scores, but it stays below approach A at top-1 on validation for every attribute except Surface texture; on test it is below or equal everywhere. **Approach A stays in use**; the fine-tuned weights are kept in `dino_finetuned_heads.pt` for a later try with more (human) labels.

2. Attribute Recall@1 / 2 / 3 (does the right label appear in the top K?)

Single-label attributes: share of test objects whose Claude label is among the K most probable classes (R@1 = top-1 accuracy). Glaze effects / defects (multi-label): mean share of an object's Claude effects found in its K most probable classes, over objects with at least one (objects labelled "none" are left out). The most-common-class guess always ranks the classes by how often they occur in training - the bar to beat.

attribute	test objects	v3 (trained) R@1 / R@2 / R@3	most-common-class guess	v2 (241 labels)	pretrained DINO alone R@1 / R@3	features chosen
Object form	238	83.6% / 89.9% / 92.0%	55.5% / 67.2% / 77.3%	79.8% / 89.1% / 94.1%	22.7% / 66.8%	dino+v1+text
Surface finish	240	45.4% / 65.8% / 75.4%	42.9% / 60.8% / 72.1%	40.8% / 57.9% / 72.5%	6.2% / 35.4%	v1
Transparency	240	74.6% / 92.1% / 98.8%	73.3% / 92.1% / 98.8%	72.5% / 92.5% / 98.8%	35.4% / 90.0%	v1
Surface texture	240	85.0% / 95.0% / 99.2%	85.0% / 95.0% / 98.3%	85.0% / 89.2% / 98.3%	4.2% / 75.4%	dino+v1
Glaze effects	240	49.4% / 66.6% / 78.2%	38.5% / 67.5% / 81.8%	–	5.7% / 33.9%	dino+v1
Possible defects	236	50.0% / 63.6% / 77.3%	27.3% / 45.5% / 100.0%	–	18.2% / 54.5%	dino+v1

Pretrained DINO alone = the best raw phrase score per class with the class-name prompt (no training) - the starting point of test B.

3. Search Recall@K (offline, leave-one-out on the 240 test objects)

Every test object is a query once; the other 239 are the gallery. A result is relevant when it shares the ticked features (Claude's labels; colour = the query's main colour within ΔE2000 20 of a region of the result). "Pick from top 3" = the user clicks the right candidate when it is shown, otherwise keeps #1 - the workflow of the new search page.

ticked features	no attributes R@3 / R@10	auto #1 R@3 / R@10	pick from top 3 R@3 / R@10	exact label typed R@3 / R@10	random R@3 / R@10
colour	100% / 100%	100% / 100%	100% / 100%	100% / 100%	94% / 99%
form	87% / 94%	90% / 93%	93% / 96%	97% / 98%	57% / 76%
surface finish	70% / 91%	64% / 82%	75% / 88%	80% / 96%	52% / 80%
transparency	86% / 98%	82% / 92%	90% / 98%	90% / 97%	83% / 95%
texture	90% / 95%	87% / 90%	92% / 95%	94% / 99%	88% / 91%
glaze effect	77% / 95%	66% / 92%	89% / 97%	93% / 100%	68% / 91%
defect	98% / 98%	96% / 98%	97% / 98%	96% / 99%	95% / 95%
colour + form	83% / 94%	88% / 92%	92% / 94%	95% / 97%	47% / 73%
colour + glaze effect	72% / 94%	66% / 87%	87% / 95%	88% / 98%	43% / 81%
colour + form + glaze effect	52% / 75%	61% / 82%	70% / 86%	71% / 89%	19% / 49%

ticked features	no attributes R@3 / R@10	auto #1 R@3 / R@10	pick from top 3 R@3 / R@10	exact label typed R@3 / R@10	random R@3 / R@10
form + surface finish + transparency	40% / 62%	47% / 72%	53% / 78%	56% / 80%	10% / 28%
colour + form + surface finish + glaze effect	33% / 50%	38% / 56%	42% / 61%	42% / 63%	7% / 20%

4. Confident answers ("validated" badge)

attribute	precision of 'answered' labels (test)	coverage	answers	98 % target met
Object form	100.0%	34.0%	81	yes
Surface finish	–	0.0%	0	no
Transparency	–	0.0%	0	no
Surface texture	96.6%	48.8%	117	no
Glaze effects	99.1%	–	860	yes
Possible defects	99.0%	–	1168	yes

An attribute is shown as validated only when its answers reached 98 % precision (Laplace-smoothed) on validation and at least 20 test answers confirm it. Uncertain attributes still give their top-3 candidates; the user decides. **Glaze effects and defects** reach 98 % only through confident *absent* decisions ("no crackle on this object"): no effect class reached 98 % for *present*, so a present effect is never auto-confirmed - it is offered as a candidate.

5. Candidates per object (all 11,930 objects)

attribute	1 candidate	2 candidates	3 candidates
Object form	9,213	1,126	1,591
Surface finish	1,401	3,402	7,127
Transparency	2,466	9,464	0
Surface texture	5,527	4,954	1,449
Glaze effects	461	1,712	9,757
Possible defects	11,735	194	1

6. Limits (stated plainly)

- All labels are Claude's visual judgement from one photo; a human check of a few hundred objects in the Tile panel would turn these numbers into verified ones (the evaluation reads human reviews automatically when they exist).
- Surface finish, transparency and texture are hard to see in one photo: at top-1 they are at or near the most-common-class guess; their value is in the top-3 list that the user picks from (search Recall@3 rises when the right one is picked).
- Defects: only 11 test objects carry a defect label, so the defect numbers are very uncertain.
- Bonus images: 379 objects of 115 photos (objects marked *not needed* left out) are in the search gallery and have top-3 attribute candidates; 375 carry their recipe's text and 374 its formula embedding (built from the Glazy export exactly as for the recipe bank). They were encoded like an uploaded photo; on 7k

objects that path matches the stored index (shape identical, appearance cosine >= 0.987, earlier image heads within 0.004 - 0.13). No Claude labels exist for them, so they are not part of any accuracy number.